Roll No._____ Name_____

DEPARTMENT OF MATHEMATICS, T. I. E. T., PATIALA

M.Sc. (Mathematics and Computing) Session: 2024-25 (Even) End Sem Examination, 2025

Time: 3 Hrs. Numerical Analysis (PMA-208) May 30, 2025 Max. Marks: 40

Note: Answer any five questions. Course Coordinator: Dr. Arvind Kumar Lal

1(a) Working with four-significant digit rounding arithmetic, find the most accurate roots of the following quadratic equation assuming its exact roots are: 92.24457002 and 0.0054204. Also, compute the relative error. (4.0) [C01, L3]

$$\frac{1}{3}x^2 - \frac{123}{4}x + \frac{1}{6} = 0.$$

(b) Perform only two iterations of the Newton's method starting with $x_0 = 1.5$ to find the real positive root of the following polynomial equation:

$x^5 - 3.7x^4 + 7.4x^3 - 10.8x^2 + 10.8x - 6.8 = 0.$ (4.0) [CO2, L3]

2(a) Suppose $f \in C[a,b]$ and $f(a)f(b) < 0$. Then bisection method produces a sequence $\{c_k\}$ approximating a root α of $f(x) = 0$. Prove that sequence $\{c_k\}$ converges to the root α. (3.5) [CO2, L2]

(b) State Gram – Schmidt Process of Orthogonalisation. In order to reduce the following matrix A into tridiagonal matrix, compute Householder matrix P_1 and $P_1 A P_1$ using Householder's transformation. (4.5) [CO3, L3]

$$A = \begin{bmatrix} 4 & 1 & -2 & 2 \\ 1 & 2 & 0 & 1 \\ -2 & 0 & 3 & -2 \\ 2 & 1 & -2 & -1 \end{bmatrix}$$

3(a) For a given second degree Newton's divided difference interpolating polynomial $P_2(x)$, derive a second degree Lagrange's interpolating polynomial. (4.0) [CO4, L2]

(b) Suppose $x_0, x_1, \cdots, x_n$ are distinct numbers in the interval $[a,b]$ and $f \in C^n[a,b]$, Then, prove that there exists a number $\xi(x)$ in (a,b) with $f[x_0, x_1, \cdots, x_n] = \frac{f^{(n)}(\xi)}{n!}$. Also, find the value of $f[1,2,3,4,5]$ if $f(x) = 2x^4 + 3x^3 + 2x^2 - x + 1$. (4.0) [CO4, L2]

4(a) Construct Hermite interpolating polynomial that fits following data and interpolate $f(2.5)$. (3.0) [CO4, L1]

x	$f(x)$	$f'(x)$
2	29	50
3	105	105

(b) Find the clamped cubic spline that passes through $(0,0)$, $(1,0.5)$, $(2,2.0)$ and $(3,1.5)$ with the boundary conditions $f'(0) = 0.2$ and $f'(3) = -1$ and also estimate $g(0.5)$. (5.0) [CO4, L4]

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5 (a)	Determine the number of subdivision N required to approximate $I = \int_0^2 \frac{dx}{4+x}$ to within 10^{-5} using composite Simpson's 1/3 rd rule and approximate the integral I using Gauss three - point formula. (4.0) [CO5, L3]
(b)	The area A inside the closed curve $y^2 + x^2 = cos\,(x)$ is given by $$A = 4\int_0^{\alpha} \sqrt{cos\,(x) - x^2}\, dx\,,$$ where α is the positive root of the equation $cos\,(x) - x^2 = 0$. (i) Working with four decimal - digits rounding arithmetic, compute α using Newton –Raphson's method starting with $x_0 = 0.8$ correct to two decimal places. (ii) Working with six - decimal digits rounding arithmetic and using Romberg's method and Trapezoidal rule to approximate area upto $O(h^4)$ only. (4.0) [CO5, L3]
6(a)	Working with four decimal digit rounding arithmetic, find the approximate solution of the following boundary value problem using finite difference method: $\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = x + 4,\ 0 \le x \le 0.6$ subject to boundary conditions: $y'(0) = 1,\ y(0.6) = 1.96$ with N = 3. (5.0) [CO5, L3]
(b)	Working with five decimal digit rounding arithmetic, estimate $y(2)$ from the initial value problem: $\frac{dy}{dx} = y + e^x,\ y(0) = 1,\ y(0.5) = 1.824,\ y(1.0) = 3.718,\ y(1.5) = 7.722$ by the Adams - Bashforth fourth order predictor - corrector method with step size 0.5. (3.0) [CO5, L3]

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